

MDWD: A Street-Level Dataset for Municipal Solid Waste Detection in Dense Urban Environments

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Abstract—Automated visual monitoring of urban environments is a growing Computer Vision research area, but municipal solid waste detection remains under-represented in dedicated benchmark resources. Existing waste-related datasets predominantly address individual litter detection, aerial imagery, or image-level classification, and none simultaneously provide street-level imagery, instance-level localization, and categorization of domestic waste streams within a structured municipal collection context. This paper introduces the Maltese Domestic Waste Dataset (MDWD), a street-level benchmark comprising 3,697 high-resolution images and 11,461 manually annotated instances across five domestic waste categories representative of Malta’s municipal collection system. The dataset captures substantial variation in location, illumination, object scale, occlusion, and urban context. To establish reproducible baselines, a cross-architecture benchmark is conducted across multiple generations of the YOLO family and a transformer-based detector. On the test set, RF-DETR-M achieves the strongest overall performance with an mAP₅₀ of 94.49% and an F1-score of 93.56%, while smaller-capacity variants maintain competitive accuracy at substantially reduced parameter counts. These results demonstrate that MDWD supports effective training of both compact CNN-based and transformer-based detectors, establishing baseline performance for future research in vision-based municipal waste monitoring.

Index Terms—Computer Vision, object detection, domestic waste dataset, waste detection, deep learning

I. INTRODUCTION

Automated visual monitoring of urban environments is an increasingly active area of Computer Vision (CV) research, addressing the operational pressures faced by municipal administrations as cities grow denser and more populous [1], [2]. Municipal solid waste (MSW) management exemplifies this pressure: in Malta, municipal waste generation reached 621 kg per capita in 2024, compared with a European Union average of 517 kg [3], while monitoring practices based on manual inspection and citizen reporting remain labor-intensive, reactive, and poorly suited to dense urban environments [4].

From a CV perspective, street-level waste monitoring constitutes a challenging instance-level object detection (OD) problem, characterized by frequent occlusion, substantial scale and illumination variation, perspective distortion from heterogeneous capture devices, visually similar waste categories, and

long-tailed class distributions [5]. Modern detection frameworks have demonstrated strong performance under such conditions [5], offering the prospect of automatically detecting and categorizing domestic waste to support monitoring, non-compliance identification, and operational decision-making [6].

Realizing this potential requires annotated datasets that reflect these conditions within an operational collection context. Existing waste-related datasets [7]–[14] predominantly address individual litter detection, aerial imagery, marine debris, or image-level recycling classification, and none simultaneously provide street-level imagery, instance-level localization, and categorization into operationally defined domestic waste streams. Malta provides a useful case study because its color-coded domestic collection scheme operates within dense, visually cluttered urban environments that exemplify broader street-level waste-monitoring challenges.

To address this gap, this paper introduces the **Maltese Domestic Waste Dataset (MDWD)**, a street-level benchmark comprising 3,697 high-resolution images and 11,461 manually annotated instances across five domestic waste categories representative of Malta’s municipal collection system. The categories correspond to distinct municipal waste streams: Mixed Waste denotes residual household refuse in black bags; Organic Waste comprises biodegradable material in small white bags collected under Malta’s organic waste scheme; Recyclable Material covers gray or green bags containing paper, cardboard, plastic, and metal packaging; Orange CMD denotes orange bags associated with Cleaning and Maintenance Division (CMD) operations; and Other Waste captures infrequently occurring items, such as bulky refuse and glass bottles, reflecting the long-tail distribution typical of real-world urban environments. To establish reproducible baselines and characterize dataset difficulty, a cross-architecture benchmark is conducted across multiple YOLO generations and the transformer-based RF-DETR framework.

The main contributions of this work are the introduction of MDWD, an operationally grounded five-class dataset aligned with real-world municipal waste collection, instance-level street-level annotations capturing the visual challenges of dense urban environments, and a reproducible cross-architecture benchmark establishing baseline performance for future research in vision-based municipal solid waste (MSW) monitoring.

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II. RELEVANT LITERATURE

This section presents the literature underpinning the proposed benchmark, covering urban monitoring applications, waste-related CV datasets, and contemporary OD paradigms.

A. Computer Vision for Urban and Environmental Monitoring

Vision-based monitoring has become a central tool for managing urban environments at scale, supporting applications ranging from infrastructure inspection to traffic and environmental assessment [1], [2], [15]. The appeal of such systems lies in their ability to convert street-level or aerial imagery into spatially structured information that can support routine operational decision-making without continuous manual inspection [15]. More recently, vision-language models (VLMs) have been proposed as a flexible alternative for urban scene interpretation, offering open-vocabulary querying without task-specific retraining [16], [17]. However, current VLM-based approaches remain largely at the exploratory or evaluation stage, and Torneiro et al. [16] explicitly identify the absence of large, annotated, task-specific datasets as a barrier to rigorous benchmarking in this space. This observation generalizes beyond urban monitoring as a whole: regardless of whether the eventual modeling paradigm is a task-specific detector or a general-purpose VLM, progress depends on the availability of annotated data that reflect the operational conditions of the target deployment environment. Within municipal waste monitoring specifically, this dependency is particularly acute, since waste accumulation exhibits considerable visual heterogeneity across geographic regions, collection schemes, and urban morphologies, limiting the transferability of models trained on visually dissimilar source data.

B. Waste-Related Computer Vision Datasets

The development of waste-related CV datasets has progressed along several largely independent trajectories, each constrained by a different combination of acquisition setting, annotation granularity, and task formulation. Table I summarizes the datasets most relevant to MDWD, while the discussion below synthesizes the limitations that motivate the present contribution.

Early efforts such as TrashNet [12] established image-level classification of recyclable materials under controlled, single-object conditions, and have been valuable for demonstrating the feasibility of transfer learning for waste classification [18], [19]. However, the absence of spatial localization and the controlled acquisition setting limit applicability to cluttered, multi-instance street scenes. TACO [10] advanced the field by introducing object-level annotations for litter detection and segmentation across a diverse set of outdoor scenes; however, its sixty fine-grained litter categories are oriented towards individual discarded items rather than the bagged, stream-categorized waste characteristic of organized municipal collection.

A second trajectory has focused on aerial and UAV-based litter detection. SODA [7], notably also collected within the Maltese context, provides bounding-box and polygon annotations

TABLE I
COMPARISON OF WASTE-RELATED CV DATASETS AND MDWD.

Dataset	Images	Classes	Annotation Type and Context
TrashNet [12]	2,527	6	Image-level classification of household litter
TACO [10]	1,500	60	Bounding boxes & polygons for street-level litter detection
GIGO [14]	25,000	5	Multilabel image classification of street-level MSW
SODA [7]	829	6	Bounding boxes & polygons of UAV litter detection
GVP [6]	5,000	2	Bounding boxes of street-level binary MSW detection
MDWD (Ours)	3,697	5	Bounding boxes of street-level multi-class MSW detection

for small-object litter detection from UAV imagery captured over garigue terrain. Bartolo et al. [8] provide a comprehensive review of this UAV-based litter detection landscape, synthesizing datasets such as BDW [13], SODA, and several ground-based collections including MJU-Waste [11], and conclude that the field requires more in-the-wild datasets exceeding several thousand annotated images. While this body of work demonstrates substantial progress in small-object and litter detection, the aerial acquisition perspective removes street-level contextual cues, such as bag color, placement pattern, and surrounding urban infrastructure, that are operationally meaningful for municipal collection workflows. BDW [13] and MJU-Waste [11] further illustrate the diversity of task formulations explored in this space, employing oriented bounding boxes for rotation-invariant bottle detection and RGB-D semantic segmentation for fine-grained object boundaries respectively; while methodologically valuable, neither addresses street-level categorization of domestic waste streams.

A third trajectory addresses street-level municipal monitoring directly. GIGO [14] provides vehicle-mounted street-level imagery with multi-label classification of urban garbage categories, offering the closest conceptual alignment with MDWD in terms of acquisition setting; however, its image-level formulation precludes instance-level localization, limiting its use for tasks requiring spatial identification of individual waste accumulations. GVP [6] addresses a comparable street-level acquisition setting through fixed-camera monitoring of garbage vulnerable points, and its binary waste/non-waste annotation scheme demonstrates the feasibility of detector-based monitoring for tracking accumulation patterns over time. Its single-class formulation, however, does not support differentiation between distinct waste streams, which is operationally important where municipal collection is organized around separate categories such as mixed, organic, and recyclable waste.

Considered collectively, none of the reviewed datasets simultaneously provide street-level imagery, instance-level bounding-box annotations, and categorization into multiple operationally defined domestic waste streams. MDWD is designed to occupy this position directly, providing street-level

imagery annotated at the instance level across five categories that mirror an operational municipal collection scheme.

C. Object Detection Architectures and Benchmarking Considerations

Selecting a detection architecture for MSW monitoring requires balancing accuracy against deployment constraints such as latency and model size. Although two-stage detectors such as Faster R-CNN achieve strong accuracy, their computational cost limits real-time deployment [5], leaving the YOLO family as the dominant real-time paradigm. YOLO11 strengthens feature extraction through spatial attention [20], YOLO12 adopts an attention-centric design while preserving real-time inference [21], and YOLO26 introduces an edge-oriented architecture with small-target-aware label assignment and a Non-Maximum Suppression (NMS)-free head [22]. Kumar et al. [6] report that YOLO11m outperforms RT-DETR, achieving an mAP_{50} of 0.91 on street-level waste imagery, supporting its inclusion as a primary CNN-based baseline. RF-DETR offers a complementary transformer-based paradigm, using neural architecture search to preserve global attention in cluttered, overlapping scenes while maintaining real-time inference [23]. Benchmarking it alongside the three YOLO generations thus spans CNN-based, attention-centric, and transformer-based paradigms under a common protocol.

Standardized, publicly available benchmarks allow comparisons to be attributed to genuine model characteristics rather than dataset-specific artifacts, enabling subsequent research to build on a shared evaluation protocol [15]. Given the absence of an existing dataset satisfying the requirements identified above, establishing such a benchmark constitutes the central motivation for MDWD.

III. METHODOLOGY

This study adopts a multi-stage methodology comprising dataset construction, annotation, data preparation, model selection, and comparative evaluation. To support reproducibility and future research, the MDWD is publicly available through Roboflow¹.

A. Dataset Construction and Annotation

MDWD comprises 3,697 high-resolution street-level images collected across Malta and Gozo between winter 2024 and late summer 2025, spanning multiple seasons and meteorological conditions to ensure temporal diversity in illumination and waste presentation patterns. Images were acquired by a third-party team using a variety of consumer-grade mobile devices, predominantly on foot with a smaller proportion captured from moving vehicles. Collection was conducted opportunistically across geographically diverse localities, with each image depicting a distinct waste pile or collection event; burst captures and video-derived frames were explicitly excluded. Images were acquired at a median resolution of 3024×4032 pixels, with an average of 3.1 annotated instances per image. Prior

¹<https://universe.roboflow.com/um-dawl-ai-lab/mdwd-maltese-domestic-waste-dataset>

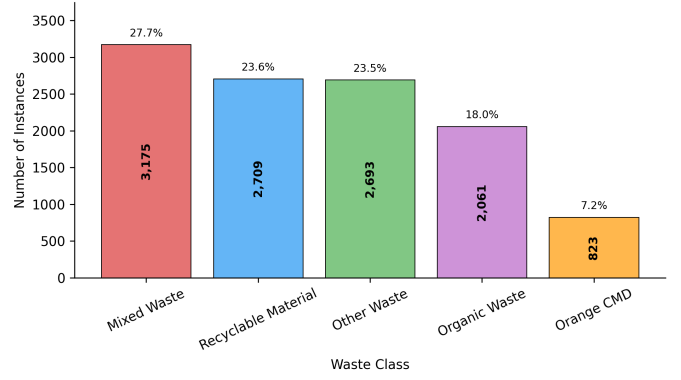


Fig. 1. Class distribution of the 11,461 annotated instances in the MDWD. The dataset exhibits moderate class imbalance, with Mixed Waste representing the largest category (27.7%) and Orange CMD the smallest (7.2%), reflecting real-world waste occurrence patterns.

to annotation, all visible personal data subject to the GDPR were manually redacted from the images [24].

All annotations were produced manually by a team of trained annotators using Roboflow exclusively as an annotation interface. The resulting annotations were subsequently reviewed and consolidated by an experienced lead annotator through a dedicated quality-assurance assessment of the complete dataset, during which bounding-box errors were corrected, category disagreements were resolved, and ambiguous instances were removed. Annotations take the form of axis-aligned bounding boxes fitted tightly to the visible extent of each instance. Objects were annotated if approximately 20% of their extent was visible; occluded regions were not extrapolated beyond their visible extent, and instances of indeterminate category were excluded. The five waste categories correspond directly to Malta’s municipal collection streams, as summarized in Figure 1 and described in Section I.

The distribution is relatively balanced across the four principal categories. Orange CMD constitutes the least represented class owing to the lower operational frequency of CMD collection activities within the surveyed areas; however, its highly distinctive visual characteristics mitigate inter-class ambiguity.

B. Dataset Preparation and Augmentation

The dataset was partitioned using an 80/10/10 train-validation-test split, corresponding to 2,958 training, 370 validation, and 369 test images. All images first underwent automatic orientation correction to remove metadata-encoded rotation artifacts, then were resized to 640×640 pixels using a stretch-based strategy consistent with the native input dimensions of the evaluated YOLO architectures. Stretch-based resizing was preferred over letterboxing to avoid padding artifacts that interact inconsistently with mosaic augmentation.

Offline augmentation was applied exclusively to the training partition via the Roboflow platform, generating ten augmented variants per image and expanding the effective training set to 29,580 images. Fixing augmentations prior to export improves reproducibility by ensuring a consistent augmented training

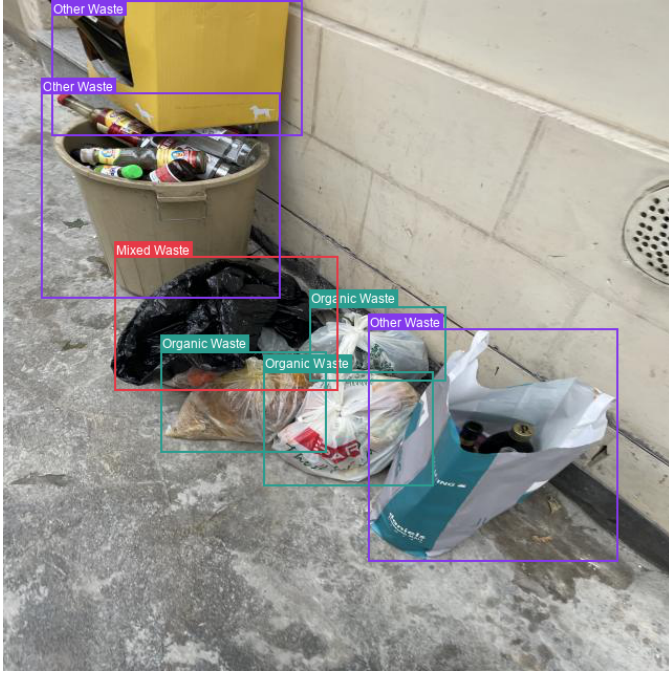


Fig. 2. Representative unaugmented MDWD sample illustrating multiple annotated waste instances across different categories, including overlapping objects, partial occlusion, and scale variation. Bounding boxes and class labels denote the ground-truth annotations.

TABLE II
OFFLINE AUGMENTATION PIPELINE APPLIED THROUGH THE ROBOFLOW PLATFORM PRIOR TO EXPORT.

Augmentation	Configuration
Resize	640 × 640 (Stretch)
Horizontal Flip	50% probability
Vertical Flip	50% probability
Hue Adjustment	±19°
Saturation Adjustment	±29%
Brightness Adjustment	±17%
Exposure Adjustment	±10%
Gaussian Blur	Up to 1 px
Motion Blur	Up to 40 px
Mosaic Augmentation	Enabled (2×2)

corpus across all evaluated models. The augmentation pipeline is summarized in Table II. Photometric transformations simulated variability in illumination arising from time of day and sensor characteristics. Horizontal and vertical flipping were retained as geometric robustness transformations. Gaussian blur of up to one pixel approximated mild optical defocus, while motion blur of up to 40 pixels simulated blur from handheld and vehicle-mounted acquisition; the latter was retained for improving validation performance. Finally, 2×2 mosaic augmentation increased contextual diversity and improved small-object detection by compositing four training images into a single input.

C. Evaluated Models and Training Configuration

To establish reproducible baselines across representative detection paradigms, three generations of the YOLO family,

TABLE III
SHARED TRAINING HYPERPARAMETERS FOR ALL LOCALLY TRAINED YOLO-FAMILY MODELS.

Parameter	Value
Image Size	640 × 640 (Stretch)
Epochs	100
Patience (Early Stopping)	10
Batch Size	32
Optimizer	AdamW
Initial Learning Rate	0.001
Weight Decay	0.0005
Momentum	0.937
Warmup Epochs	3
Seed	42

namely YOLO11, YOLO12, and YOLO26, as well as the transformer-based RF-DETR framework were evaluated, motivated by the demonstrated effectiveness of modern OD architectures for waste and litter detection [8]. YOLO11, YOLO12, and RF-DETR were evaluated at nano (N), small (S), and medium (M) variants; YOLO26 was additionally evaluated at the large (L) variant to provide a more extensive capacity analysis of the most recent generation. Detector architecture and model scale constitute the principal variables of interest; other experimental conditions were held constant to the extent permitted by the respective training environments.

All YOLO experiments were conducted using the Ultralytics framework (v8.4.60), PyTorch 2.10.0, and CUDA 13.0 on an NVIDIA GeForce RTX 4090 with 24 GB of device memory. Models were initialized from publicly available COCO pre-trained checkpoints using a fixed random seed of 42 and deterministic execution for reproducibility. The shared hyperparameter configuration is presented in Table III, and all runs were tracked via Weights & Biases.

RF-DETR experiments were conducted via the Roboflow cloud training platform, with models initialized from Objects365 pre-trained weights and trained for 100 epochs under the same early stopping criterion; all remaining hyperparameters were set to Roboflow platform defaults. While both families were trained on the same offline-augmented dataset for the same number of epochs, differences in pre-training source and infrastructure mean that cross-architecture comparisons should be interpreted as a pragmatically motivated benchmark reflecting realistic model-development conditions, rather than as a fully controlled ablation. This is consistent with this work’s primary contribution: the introduction and characterization of MDWD as a benchmark dataset.

IV. EVALUATION

This section presents the comparative evaluation of all benchmark architectures on MDWD, reporting performance on both the validation and test partitions, with test-set results constituting the primary basis for comparison. Performance is assessed using mAP_{50} , $mAP_{50:95}$, precision, recall, and F1-score.

TABLE IV
CROSS-ARCHITECTURE BENCHMARK RESULTS ON THE MDWD, REPORTING VALIDATION (V) AND TEST (T) PERFORMANCE FOR EACH METRIC. RF-DETR-M ACHIEVED THE STRONGEST OVERALL PERFORMANCE AMONG THE MEDIUM-SIZED MODELS AND THE HIGHEST TEST-SET mAP_{50} , $mAP_{50:95}$, RECALL, AND F1-SCORE.

Model	Params (M)	Size (MB)	mAP ₅₀ (%)		mAP _{50:95} (%)		Precision (%)		Recall (%)		F1-score (%)	
			V	T	V	T	V	T	V	T	V	T
YOLO26-N	2.51	5.14	86.98	86.59	69.84	68.86	92.23	93.62	80.37	79.06	85.89	85.73
YOLO26-S	9.95	19.38	89.99	90.14	75.60	75.27	94.70	94.89	84.87	86.19	89.51	90.33
YOLO26-M	21.78	42.00	90.74	89.51	76.21	75.97	94.06	93.07	86.14	86.25	89.92	89.53
YOLO26-L	26.18	50.54	91.52	89.76	79.51	78.20	95.78	97.18	87.32	86.98	91.36	91.80
YOLO12-N	2.57	5.26	88.35	88.01	71.22	70.29	91.55	94.02	82.16	80.39	86.60	86.67
YOLO12-S	9.26	18.06	89.41	90.00	74.14	74.41	92.69	95.72	84.58	85.13	88.45	90.11
YOLO12-M	20.14	38.88	91.47	90.30	76.56	75.84	94.16	95.72	87.59	85.73	90.76	90.45
YOLO11-N	2.59	5.22	86.36	85.71	68.88	67.13	90.36	92.56	80.37	78.08	85.07	84.71
YOLO11-S	9.43	18.29	89.74	90.06	73.46	73.34	93.45	94.24	83.50	85.14	88.20	89.46
YOLO11-M	20.06	38.64	88.67	89.72	73.54	74.77	92.03	94.19	82.00	85.68	86.73	89.73
RF-DETR-N	30.16	115.24	91.92	90.21	72.34	70.88	95.15	95.17	87.87	85.53	91.37	90.10
RF-DETR-S	31.81	121.52	94.27	92.61	77.38	76.17	95.87	96.39	91.04	89.24	93.40	92.68
RF-DETR-M	33.38	127.54	95.81	94.49	80.04	78.21	95.53	96.63	91.70	90.69	93.57	93.56

Note: The best **validation** result for each metric is highlighted in light red, while the best **test** result is highlighted in light blue.

A. Cross-Architecture Benchmark Results

Table IV reports validation and test performance across all evaluated architectures. Precision, recall, and F1-score are computed using fixed confidence and IoU thresholds of 0.50, while mAP_{50} and $mAP_{50:95}$ follow each framework’s native COCO-style evaluation. RF-DETR-M achieves the strongest overall test-set performance among the medium-sized models, with RF-DETR-S remaining closely competitive despite its smaller footprint. Within the YOLO family, YOLO26-L performs best overall, while smaller variants such as YOLO12-S and YOLO26-S remain competitive with substantially fewer parameters. All models achieve test-set F1-scores above 84%, demonstrating that MDWD supports effective training across diverse architectures and model capacities. The small validation-to-test differences further indicate stable performance on held-out imagery.

B. Class-Level Analysis

Class-level results for RF-DETR-M reveal consistently strong performance across most categories on the test set. Recyclable Material achieves the highest performance (mAP_{50} 98.3%), followed by Organic Waste and Mixed Waste. Orange CMD exhibits high precision (96.9%) but the lowest recall among all categories (79.5%), indicating conservative predictions and a tendency to miss true instances. Other Waste is the most challenging category overall, with the lowest mAP_{50} (93.2%) and the highest false negative count (46), though its recall (83.5%) remains higher than Orange CMD’s, consistent with its heterogeneous, infrequently occurring waste types and greater intra-class variability.

C. Discussion

The strong performance achieved across convolutional, attention-centric, and transformer-based architectures indicates

that MDWD is both learnable and sufficiently well annotated to support training across diverse detection paradigms. At the same time, the reduced performance observed for Orange CMD and particularly Other Waste demonstrates that the dataset remains challenging and provides meaningful headroom for future research on long-tailed and heterogeneous waste categories. The consistency between validation and test results further supports the reliability of the benchmark as a representative measure of generalization to unseen imagery. These results establish a reproducible baseline for future research on MDWD rather than asserting superiority of any individual architecture.

V. CONCLUSION

This paper introduced MDWD, a street-level MSW detection benchmark comprising 3,697 images and 11,461 manually annotated instances across five operationally defined waste categories. By combining street-level imagery with instance-level annotations aligned with Malta’s municipal collection scheme, MDWD addresses a gap in existing waste-related datasets and establishes reproducible baselines for multi-class municipal waste detection. Evaluation across CNN- and transformer-based architectures confirmed the dataset’s suitability for detector training, with RF-DETR-M achieving the strongest overall test-set performance.

Although geographically centered on the Maltese archipelago and currently limited to bounding-box detection, MDWD provides a strong foundation for further research. Future work may investigate cross-region generalization, finer-grained waste taxonomies, promptable segmentation models such as SAM 3 [25], self-supervised visual representations such as V-JEPA 2 [26], and saliency-based prioritization mechanisms [27], supporting more robust scene understanding and operational municipal monitoring.

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